



HAM LAB Process Calculator :

Technical Documentation

Version: 2.2

Framework: Python 3.x (CustomTkinter, Matplotlib, FPDF)

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1. Module A: Wire Arc Additive Manufacturing (WAAM)

This module calculates the thermal and mass deposition physics for arc-based additive manufacturing. It employs a **reverse-calculation algorithm**: instead of predicting geometry from parameters, it calculates the required Wire Feed Speed (WFS) and Energy input needed to achieve a specific *Target Bead Geometry*.

1.1 Inputs Definition

Symbol	Parameter	Unit	Description
V	Voltage	Volts (V)	The electrical potential difference across the arc.
I	Current	Amperes (A)	The flow of electric charge through the arc.
η	Efficiency	Unitless	Process thermal efficiency factor (e.g., 0.8 for GMAW/MIG), accounting for radiative and convective heat losses.
S	Travel Speed	mm/min	The velocity of the torch relative to the substrate.

d_w	Wire Diameter	mm	The diameter of the feedstock wire.
w_b	Target Bead Width	mm	The desired width of the deposited bead.
h_l	Layer Height	mm	The target Z-increment per layer (slicer height).
L_{bead}	Bead Length	mm	The length of the deposition path.
ρ	Material Density	g/cm ³	Density of the feedstock material (Standard Steel: 7.85 g/cm ³).

1.2 Assumptions & Constraints

1. **Parabolic Bead Profile:** The cross-section of the deposited bead is assumed to be parabolic. A shape factor of **0.75** is used to relate the bounding box ($w_b \times h_l$) to the actual cross-sectional area.
2. **Mass Conservation:** The volume of wire consumed equals the volume of the bead deposited (spatter losses are considered negligible for this calculation).
3. **Steady State:** Voltage, Current, and Travel Speed are assumed constant throughout the layer deposition.

1.3 Mathematical Model & Output Definitions

A. Power & Energy Metrics

- **Total Input Power (P_{total}):** Gross electrical power.

$$P_{total} = V \times I \quad [Watts]$$

- **Effective Power (P_{eff}):** The actual thermal energy transferred to the substrate.

$$P_{eff} = P_{total} \times \eta \quad [Watts]$$

- **Linear Energy Density (E_L):** Critical for determining heat affected zone (HAZ) size and microstructure.

$$E_L = \frac{P_{eff}}{S/60} \quad [J/mm]$$

- **Single Layer Energy (E_{layer}):** Total energy required to deposit one pass.

$$E_{layer} = E_L \times L_{bead} \quad [Joules]$$

B. Mass & Geometry Calculations

- **Bead Cross-Sectional Area (A_{bead}):** Derived from the parabolic assumption.

$$A_{bead} = \frac{w_b \times h_l}{0.75} \quad [mm^2]$$

- **Volumetric Deposition Rate (V_{rate}):** Volume of material deposited per unit time.

$$V_{rate} = A_{bead} \times S \quad [mm^3/min]$$

- **Calculated Wire Feed Speed (WFS):** The control parameter required to supply the necessary volume.

$$WFS = \frac{V_{rate}}{\pi \cdot (d_w/2)^2} \times \frac{1}{1000} \quad [m/min]$$

- **Mass Deposition Rate:**

$$M_{rate} = V_{rate} \times \rho \times 10^{-6} \times 60 \quad [kg/h]$$

2. Module B: PM Consolidation (Powder Metallurgy)

This module simulates the **Solid-State Consolidation** of metal powder via uni-axial pressing. It models the iterative densification process, accounting for strain hardening of the compact and geometric losses (flash) per layer.

2.1 Inputs Definition

Symbol	Parameter	Unit	Description
L, W, E	Target Dimensions	mm	The final dimensions of the consolidated block.
CR	Compression Ratio	Unitless	Ratio of Loose Powder Height to Green Compact Height (typically 2.0–2.5).
P_{base}	Base Pressure	MPa	Initial hydraulic pressure applied to the first layer.
α	Resistance Factor	Unitless	Strain Hardening Coefficient: Represents the increase in pressure required to densify subsequent layers as the material hardens (e.g., 0.03 = 3% increase per layer).
β	Distortion Factor	Unitless	Geometric Loss Coefficient: Represents the reduction in effective pressure area due to flash formation or edge effects (e.g., 0.05 = 5% area loss per layer).
h_{del}	Deposition Height	mm	The height of loose powder added per

			cycle.
h_{mach}	Machining Cut	mm	Top surface material removed per cycle (simulating milling).

2.2 Assumptions & Constraints

1. **Iterative Hardening:** The material is assumed to work-harden linearly with each cycle, requiring progressively higher pressure to achieve the same density.
2. **Geometric Decay:** The effective load-bearing area is assumed to decrease slightly with each layer due to geometric distortion (flash) at the die edges.
3. **Constant Stroke:** The hydraulic press stroke is constant, determined by the fill height and compression ratio.

2.3 Mathematical Model (Layer-wise Iteration)

The simulation iterates from layer $n = 1$ to N :

A. Effective Area (A_n) Simulates the reduction in effective pressure area due to distortion/flash (β).

$$A_n = A_{n-1} \times (1 - \beta) \quad [mm^2]$$

(Initial Condition: $A_1 = L \times W$)

B. Required Pressure (P_n) Simulates the material resistance increasing due to work hardening (α).

$$P_n = P_{n-1} \times (1 + \alpha) \quad [MPa]$$

C. Peak Force (F_n) The hydraulic force required for layer n .

$$F_n = A_n \times P_n \times 10^{-3} \quad [kN]$$

D. Stroke & Energy (W_n)

- **Compression Stroke (δ)**: Distance the ram travels to compress loose powder to solid.

$$\delta = h_{dep} \times (CR - 1) \quad [mm]$$

- **Work Done ($Energy_n$)**: Energy consumed for consolidation.

$$Energy_n = F_n \times \delta \quad [Joules]$$

3. Module C: Friction Stir Welding (FSW)

This module estimates the torque, heat generation, and forces involved in **Friction Stir Welding**, a solid-state joining process.

3.1 Inputs Definition

Symbol	Parameter	Unit	Description
ω	Rotation Speed	RPM	Angular velocity of the FSW tool.
v	Weld Speed	mm/s	Traverse speed of the tool along the joint line.
F_z	Downward Force	kN	Axial force applied by the tool shoulder.
D_s	Shoulder Diameter	mm	Diameter of the tool shoulder (primary heat source).
μ	Friction Coefficient	Unitless	Dynamic friction coefficient between tool and workpiece.

3.2 Assumptions

1. **Shoulder Dominance:** Heat generation is assumed to be dominated by the shoulder contact (approx. 80–90% of total heat), with pin contribution considered negligible for this estimation.
2. **Contact Friction:** The torque is modeled using a sliding friction model, modified by a thermal softening factor at high RPMs.

3.3 Mathematical Model & Output Definitions

A. Torque (T)

Estimated using the Analytical Contact Friction Model. A **Softening Factor** (λ) is applied to prevent unrealistic torque values at high RPMs where material softens.

$$\lambda_{soft} = \max(0.5, 1.0 - \frac{RPM}{3500})$$

$$T \approx \frac{2}{3} \times \mu \times F_z \times \frac{D_s}{2} \times \lambda_{soft} \quad [N \cdot m]$$

B. Heat Generation (Q)

Mechanical power converted to thermal energy.

$$Q = T \times \omega_{rad/s} = T \times \left(\frac{2\pi \times RPM}{60} \right) \quad [Watts]$$

C. Specific Energy (E_s)

Energy input per unit length of weld.

$$E_s = \frac{Q}{v} \quad [J/mm]$$

4. Comparative Analysis Logic

The System generates a side-by-side comparison (Module 4) to normalize the energy and force requirements of WAAM vs. PM for a **Standardized Target Volume** (

$$Vol = L \times W \times H).$$

4.1 Normalization Algorithms

- **WAAM Normalization:**

- Calculates total path length to fill Vol based on bead geometry.
- $Total\ Energy = (Linear\ Energy) \times (Total\ Path\ Length)$.
- $Peak\ Force = 0\ kN$ (Non-contact process).

- **PM Normalization:**

- Calculates number of layers to fill H .
- $Total\ Energy = \sum (Work\ per\ layer)$.
- $Peak\ Force = Max(Force_{layer})$.

4.2 Comparison Outputs

The generated report contrasts:

1. **Energy Efficiency:** Ratio of total Joules consumed. WAAM is typically higher (fusion) vs. PM (solid-state).
2. **Productivity:** Ratio of total process time (Arc-on time vs. Press cycle time).
3. **Mechanical Load:** Peak reaction forces required by the machinery (0 kN for WAAM vs >100 kN for PM).

5. References

The mathematical models and physics assumptions implemented in this controller are derived from the following standard engineering texts and research papers:

1. **WAAM Energy & Geometry:**

- Williams, S. W., et al. (2016). "Wire + Arc Additive Manufacturing." *Materials Science and Technology*, 32(7), 641-647.
- Suryakumar, S., et al. (2011). "Weld bead modeling and process optimization in Hybrid Layered Manufacturing." *Computer-Aided Design*, 43(4), 331-344. (Source of the Parabolic Shape Factor assumption).

2. Powder Metallurgy (Consolidation Physics):

- German, R. M. (2013). "Powder Metallurgy and Particulate Materials Processing." *Metal Powder Industries Federation*. (Heckel Equation variations for pressure-density relationships).

3. Friction Stir Welding (Torque & Heat):

- Mishra, R. S., & Ma, Z. Y. (2005). "Friction stir welding and processing." *Materials Science and Engineering: R: Reports*, 50(1-2), 1-78.
- Schmidt, H., & Hattel, J. (2005). "A local model for the thermomechanical conditions in friction stir welding." *Modelling and Simulation in Materials Science and Engineering*. (Source of the analytical torque estimation model: $T = \frac{2}{3}\mu FR$).

End of Documentation

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